A Framework for Evaluating Neural Network Deployability on Analog In-Memory Computing Hardware

Abstract

Analog in-memory computing (AIMC) promises 10–1000x inference energy reduction, but its software ecosystem is fragmented across device, circuit, and architecture-search levels (e.g., CrossSim, IBM's AIHWKit, NeuroSim, AnalogNAS-Bench), and existing frameworks do not address the question for hardware-algorithm codesigners: which neural computations can harness analog substrates effectively and which need substantial digital support? We study this across seven modern architecture families: Transformers, Neural ODEs, S4D-style state-space models, Deep Equilibrium Networks, Normalizing Flows, Hopfield-style Energy-Based Models, and Diffusion. Our framework decomposes PyTorch models into a typed taxonomy of analog circuit-level primitives (crossbar MVM, op-amp integrator, RC decay, ADC/DAC boundary, among 20+ others) labeled analog, digital, or hybrid; injects physics-grounded conductance mismatch, kT/C thermal noise, and ADC quantization; reports per-architecture analog amenability scores and energy/latency estimates against digital baselines; and compiles each family to a JAX-based analog circuit IR for simulation. In a pilot study on small models, three failure modes emerge: single-pass architectures remain robust; iterative fixed-point solvers fail under compounded mismatch; and multi-step samplers accumulate quantization error even at zero fabrication mismatch. A 14-model CIFAR-10 and WikiText-2 benchmark infrastructure is implemented, with full-scale runs planned as resources become available.

Framework

The framework is a two-layer profiling stack separating circuit physics from architectural semantics. Layer 1 simulates crossbar MVMs with conductance mismatch and Johnson–Nyquist noise, and models representative analog realizations for these families: op-amp integrators with kT/C noise for ODEs and flows, and Hopfield-style feedback for DEQs. Layer 2 builds a typed IR whose operators are labeled ANALOG, DIGITAL, or HYBRID (computational semantics) and assigned a TargetBackend for analog circuit deployment. Each node carries analog compute fraction, D/A boundary count, and a dynamics-aware amenability score. Energy and latency are estimated via a HardwareProfile and reported alongside accuracy in every sweep. Each IR node carries a Domain label and TargetBackend selector, and each model family exports to analog circuit schedules. Sweeps default to circuit-mode evaluation (RC-integrator, Hopfield, critically damped Langevin), so degradation reflects analog substrate behavior rather than digital approximations. A simple PyTorch API converts trained models into this analog-annotated IR to drive Monte Carlo sweeps and energy/latency reports.

Pilot Results

A pilot study evaluates all families under 0–15% conductance mismatch with 50 Monte Carlo trials per setting. Five single-pass families (Transformer, Neural ODE, S4D, Normalizing Flow, Hopfield-style EBM) remained above the pilot robustness threshold across tested mismatch levels, while our DEQ variants lost convergence around 12% mismatch, as fixed-point iteration repeatedly applies a noisy map. Diffusion exhibits a structural failure: in our pilot setup diffusion quality stayed below the target threshold even without fabrication mismatch under conservative ADC placement because quantization accumulates across a 20-step sampling pipeline, whereas thermal noise is negligible throughout. A benchmark infrastructure for larger, real-world tasks (CIFAR-10 and WikiText-2) is implemented, with training runs and sweeps planned.

Significance and Status

Recent AIMC silicon (e.g., IBM PCM-based AIMC chips and gain-cell self-attention accelerators) delivers real efficiency gains, but adoption is gated by which models actually tolerate device nonidealities. Prior tools target device, circuit, or architecture-search levels (e.g., CrossSim, AIHWKit, NeuroSim, DNN+NeuroSim, CiMLoop, AnalogNAS-Bench); our work instead focuses on the architecture-family question with a typed IR, circuit-mode evaluation, energy/latency modeling, and a compilation bridge from PyTorch graphs to analog circuits. The results suggest that computational structure, not device physics alone, determines analog deployability: single-pass architectures with non-iterative inference paths are the most promising AIMC target, while iterative fixed-point and multi-step sampling pipelines expose repeated D/A conversions and amplify nonidealities.